THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In an AP with $a = 8$ and $d = -2$, the 2nd term is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 6

2. The 1st term of an AP is 7 and the 3rd term is 17. The common difference is: 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 7

3. The common difference of the AP $7, 12, 17, 22, \ldots$ is: 1

(a) 5 **(b)** -5 **(c)** 10 **(d)** 6

4. The 6th term of the AP $2, 9, 16, 23, \ldots$ is: 1

(a) 39 **(b)** 36 **(c)** 37 **(d)** 38

5. The sum of the first 6 terms of $5, 10, 15, 20, \ldots$ is: 1

(a) 107 **(b)** 105 **(c)** 104 **(d)** 106

6. Is $1, 6, 11, 16, \ldots$ an AP? 1

(a) Yes **(b)** Only if $d > 0$ **(c)** Cannot say **(d)** No

7. In an AP with $a = 11$ and $d = 4$, the 2nd term is: 1

(a) 14 **(b)** 16 **(c)** 15 **(d)** 17

8. The 1st term of an AP is 2 and the 3rd term is 14. The common difference is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 5

9. The common difference of the AP $-2, 4, 10, 16, \ldots$ is: 1

(a) 6 **(b)** -6 **(c)** 12 **(d)** 7

10. The 5th term of the AP $10, 15, 20, 25, \ldots$ is: 1

(a) 32 **(b)** 31 **(c)** 30 **(d)** 29

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the common difference of the AP $20, 17, 14, 11, \ldots$ 2

12. Is $15, 11, 7, 3, \ldots$ an AP? If so, find the next term. 2

13. Write the first four terms of the AP with $a = 7$ and $d = 4$. 2

14. Find the common difference of the AP $4, \frac{11}{2}, 7, \frac{17}{2}, \ldots$ 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the 8th term of the AP $2, 9, 16, 23, \ldots$ — 3

16. Find the sum of the first 12 terms of the AP $1, 5, 9, 13, \ldots$ — 3

17. Which term of the AP $2, 6, 10, 14, \ldots$ is 54? — 3

OR

Find the 15th term of the AP $10, 15, 20, 25, \ldots$ — 3

18. Find the sum of the first 15 terms of the AP $7, 9, 11, 13, \ldots$ — 3

OR

Which term of the AP $10, 13, 16, 19, \ldots$ is 31? — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms are there in the AP 2, 8, 14, \dots, 68? — 5

OR

In an AP, $a = 2$, $d = 5$, $n = 18$. Find a_n and S_n. — 5

20. How many terms are there in the AP 10, 14, 18, \dots, 106? — 5

OR

In an AP, $a = 15$, $d = 2$, $n = 25$. Find a_n and S_n. — 5

END OF QUESTION PAPER — Total: 40 Marks

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